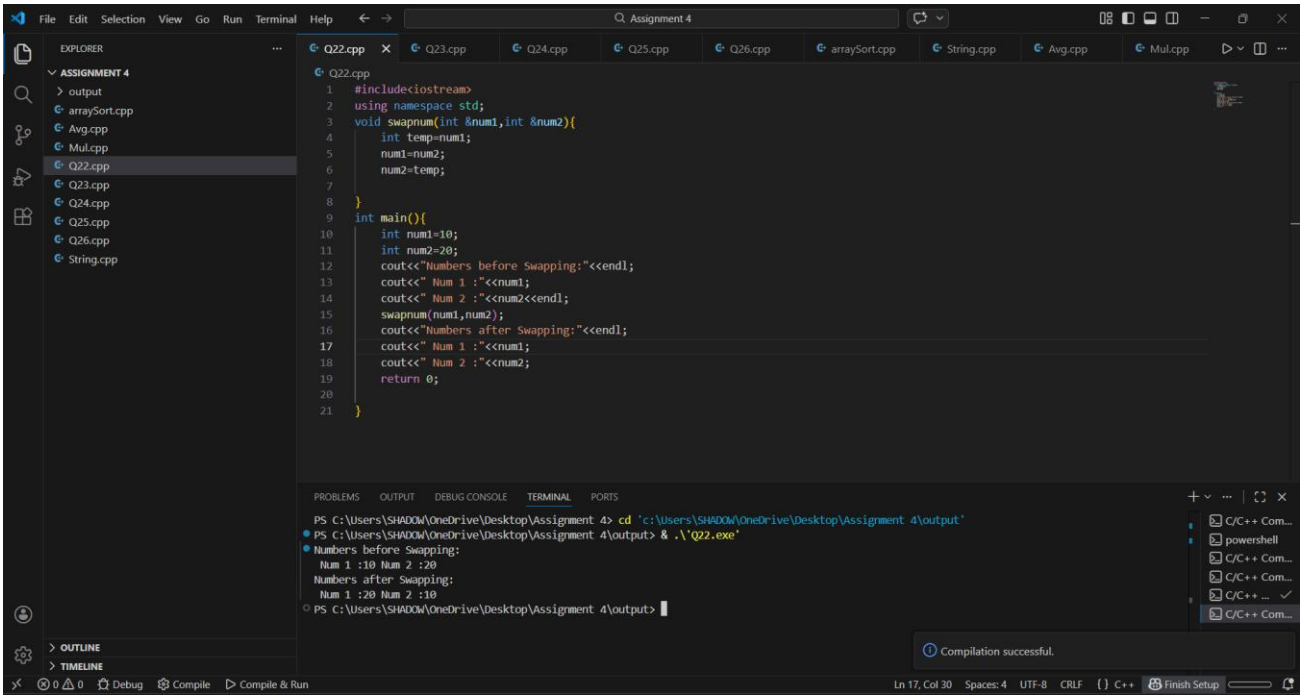
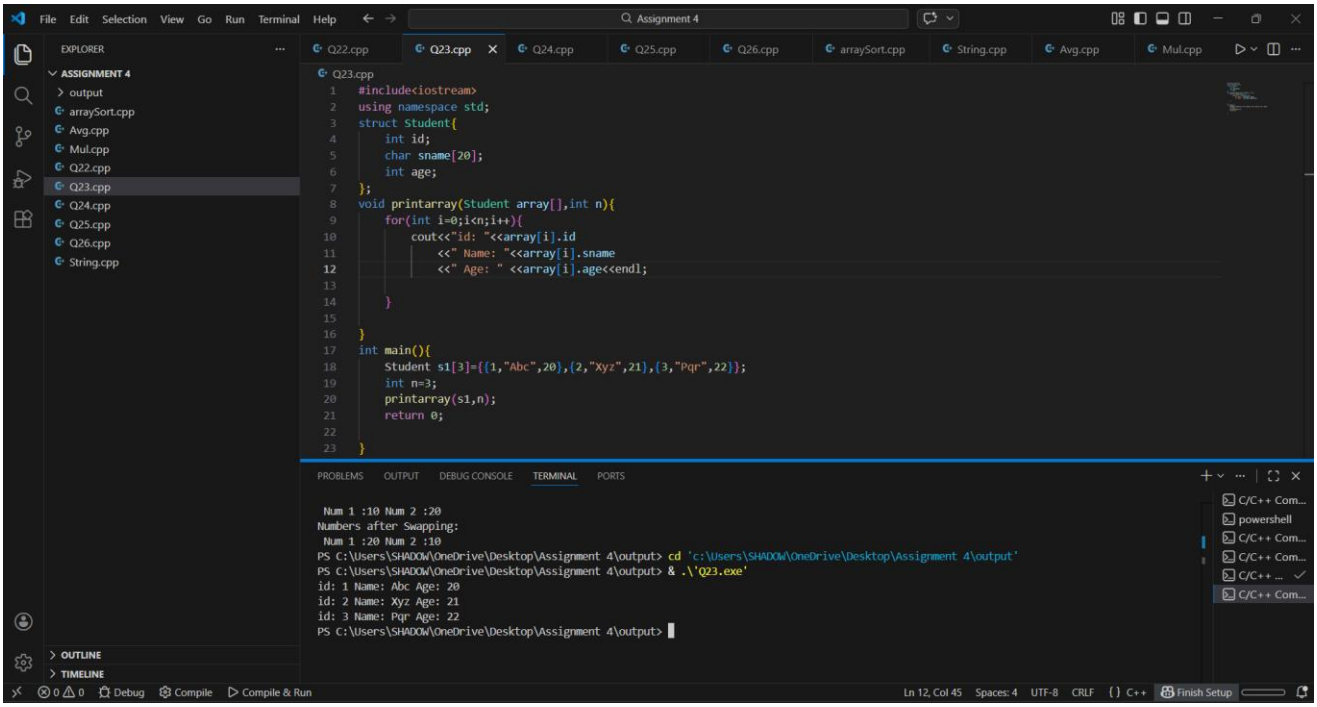
Declare function and call it by reference for swapping numbers.

```
#include<iostream>
using namespace std;
void swapnum(int &num1,int &num2){
   int temp=num1;
   num1=num2;
   num2=temp;

}
int main(){
   int num1=10;
   int num2=20;
   cout<<"Numbers before Swapping:"<<endl;
   cout<<" Num 1 :"<<num1;
   cout<<" Num 2 :"<<num2<<endl;
   swapnum(num1,num2);
   cout<<"Numbers after Swapping:"<<endl;
   cout<<" Num 1 :"<<num1;
   cout<<" Num 2 :"<<num2;
   return 0;

}
```



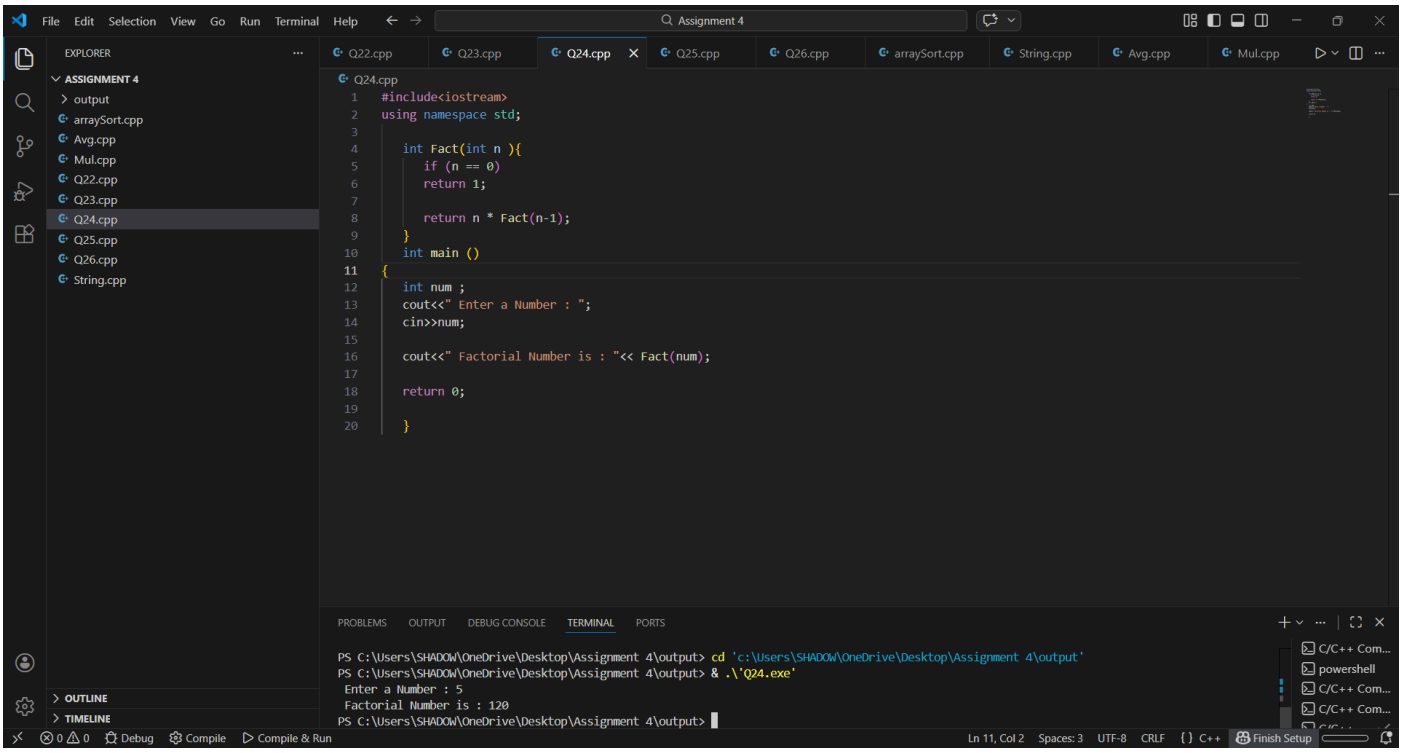
Create function for printing an array and call it from main (Student array).

```
#include<iostream>
using namespace std;
struct Student{
  int id;
  char sname[20];
  int age;
};
void printarray(Student array[],int n){
  for(int i=0;i<n;i++){
    cout<<"id: "<<array[i].id
      <<" Name: "<<array[i].sname
      <<" Age: " <<array[i].age<<endl;
  }
}
int main(){
  Student s1[3]={{1,"Abc",20},{2,"Xyz",21},{3,"Pqr",22}};
  int n=3;
  printarray(s1,n);
  return 0;

}
```



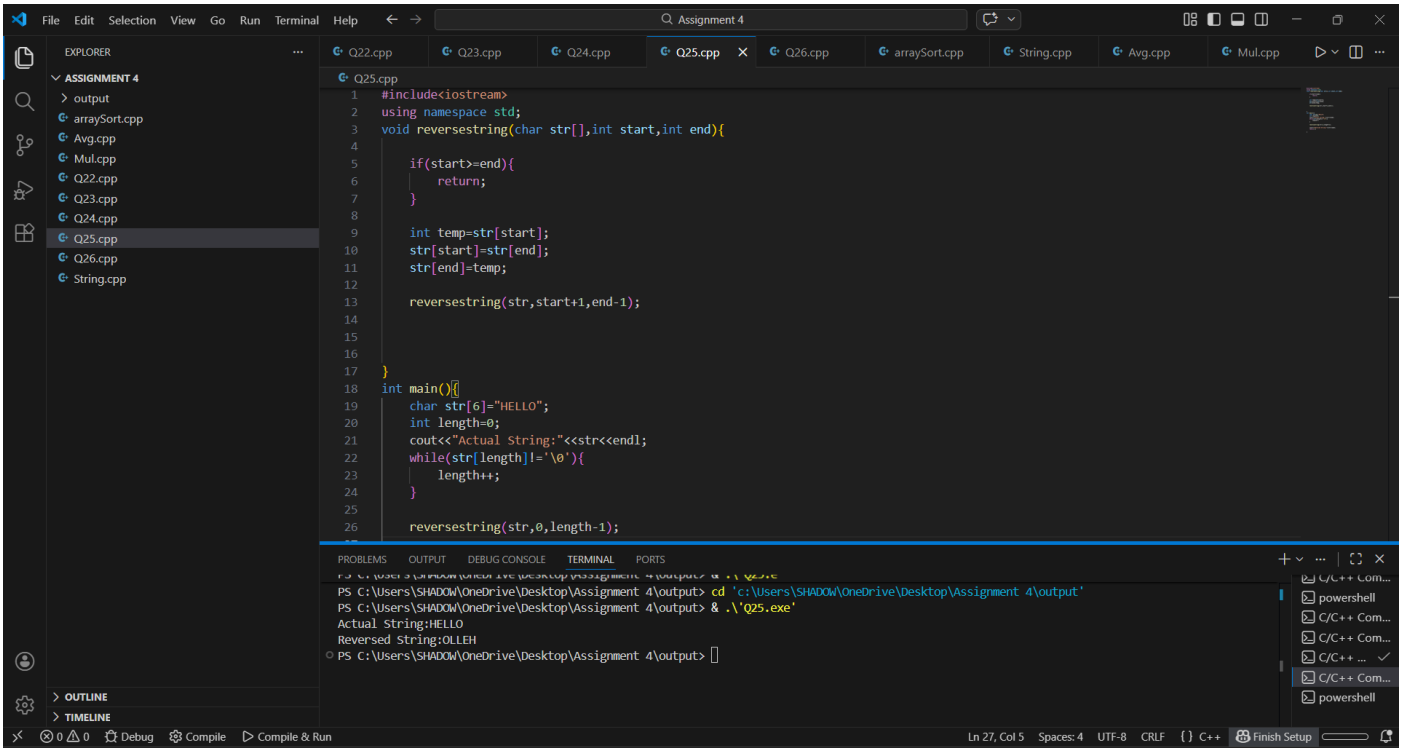
Function for print factorial series with recursive.

```
#include<iostream>

using namespace std;

  int Fact(int n ){

    if (n == 0)

    return 1;


    return n * Fact(n-1);

  }

  int main ()

{

  int num ;

  cout<<" Enter a Number : ";

  cin>>num;

  cout<<" Factorial Number is : "<< Fact(num);

  return 0;

  }
```



Reverse string (char array) with recursion.

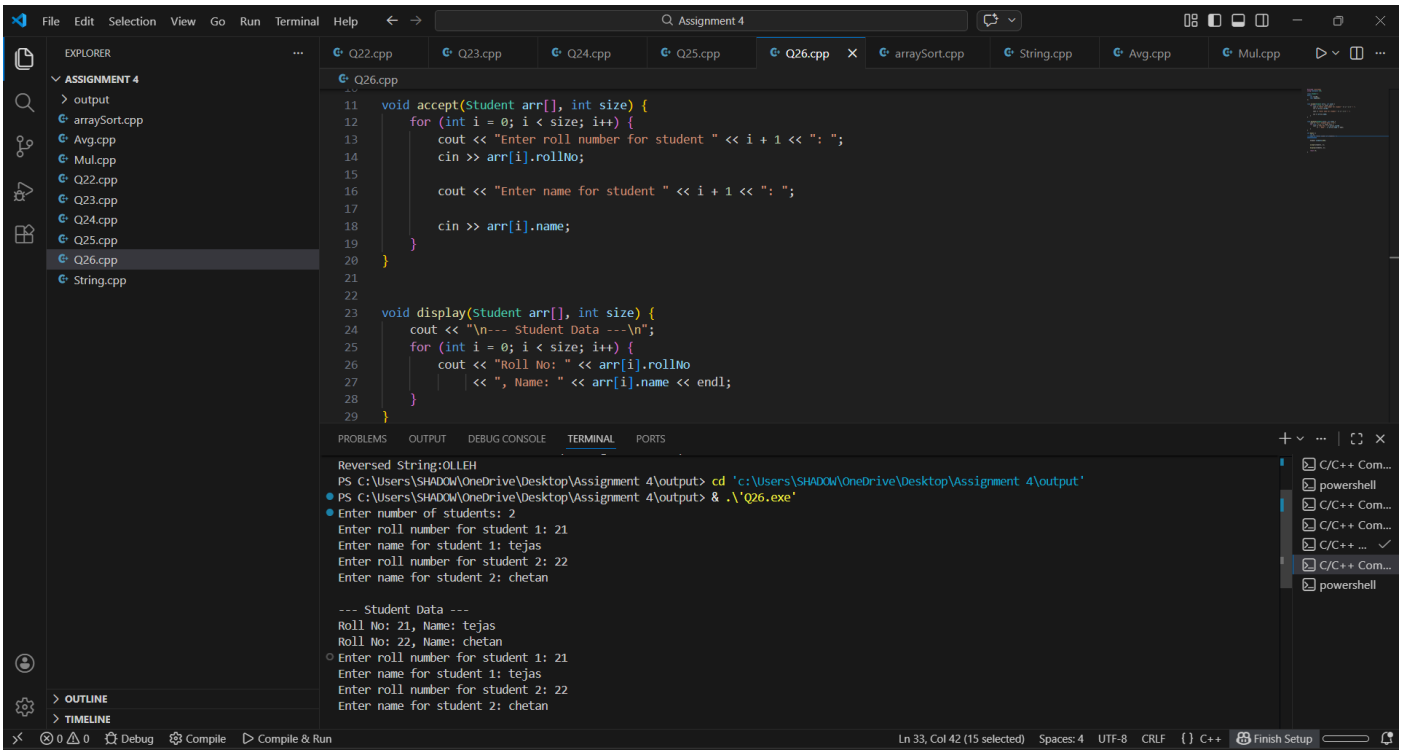
```
#include<iostream>
using namespace std;
void reversestring(char str[],int start,int end){

   if(start>=end){
      return;
   }

   int temp=str[start];
   str[start]=str[end];
   str[end]=temp;

   reversestring(str,start+1,end-1);



}
int main(){
   char str[6]="HELLO";
   int length=0;
   cout<<"Actual String:"<<str<<endl;
   while(str[length]!='\0'){
      length++;
   }

   reversestring(str,0,length-1);

   cout<<"Reversed String:"<<str<<endl;
   return 0;

}
```



Create Student class and add rollNo and name in it. Create Accept() func-tion which will take an array of Students as input. Write one more func-tion to display student data. This Display() function should receive Stu-dent array.

```
#include <iostream>

using namespace std;


class Student {

public:

  int rollNo;

  char name[50];

};


void accept(Student arr[], int size) {

  for (int i = 0; i < size; i++) {

    cout << "Enter roll number for student " << i + 1 << ": ";

    cin >> arr[i].rollNo;


    cout << "Enter name for student " << i + 1 << ": ";


    cin >> arr[i].name;

  }
```

```
}

void display(Student arr[], int size) {
  cout << "\n--- Student Data ---\n";
  for (int i = 0; i < size; i++) {
    cout << "Roll No: " << arr[i].rollNo
        << ", Name: " << arr[i].name << endl;
  }
}

int main() {
  int n;
  cout << "Enter number of students: ";
  cin >> n;
  Student students[100];
  accept(students, n);
  display(students, n);
  return 0;
}
```

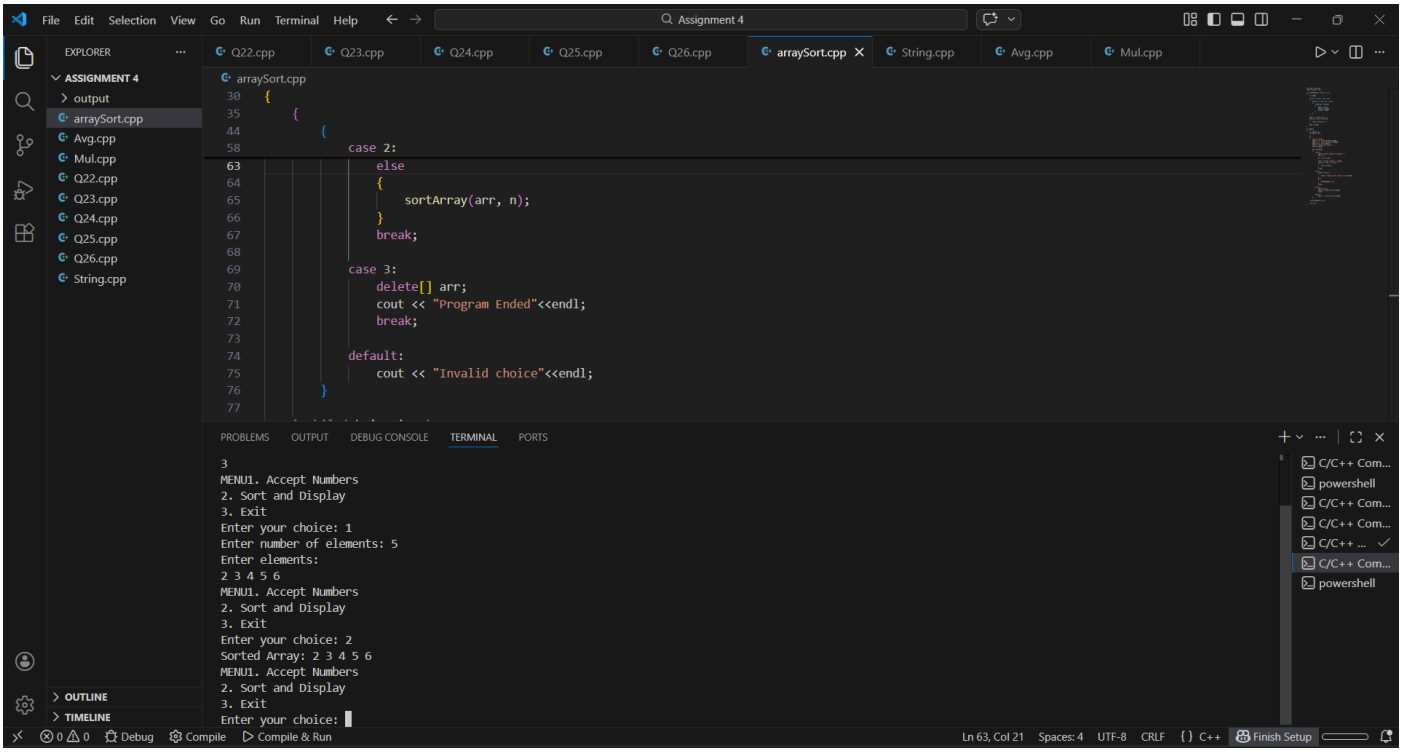


Menus: 1. Accept n numbers from users and store them in to dynamically allocated array. Write a function to sort and display sorted array.

```
#include <iostream>

using namespace std;


void sortArray(int arr[], int n)

{

  int temp;


  for(int i = 0; i < n-1; i++)

  {

    for(int j = i+1; j < n; j++)

    {

      if(arr[i] > arr[j])

      {

        temp = arr[i];

        arr[i] = arr[j];

        arr[j] = temp;

      }

    }

  }


  cout << "Sorted Array: ";

  for(int i = 0; i < n; i++)

  {

    cout << arr[i] << " ";

  }

  cout << endl;

}


int main()

{

  int choice, n;

  int *arr = NULL;
```

```
do
{
  cout << "MENU";
  cout << "1. Accept Numbers"<<endl;
  cout << "2. Sort and Display"<<endl;
  cout << "3. Exit"<<endl;
  cout << "Enter your choice: ";
  cin >> choice;

  switch(choice)
  {
    case 1:
      cout << "Enter number of elements: ";
      cin >> n;

      arr = new int[n];

      cout << "Enter elements: "<<endl;
      for(int i = 0; i < n; i++)
      {
        cin >> arr[i];
      }
      break;

    case 2:
      if(arr == NULL)
      {
        cout << "Please enter numbers first"<<endl;
      }
      else
      {
        sortArray(arr, n);
      }
```

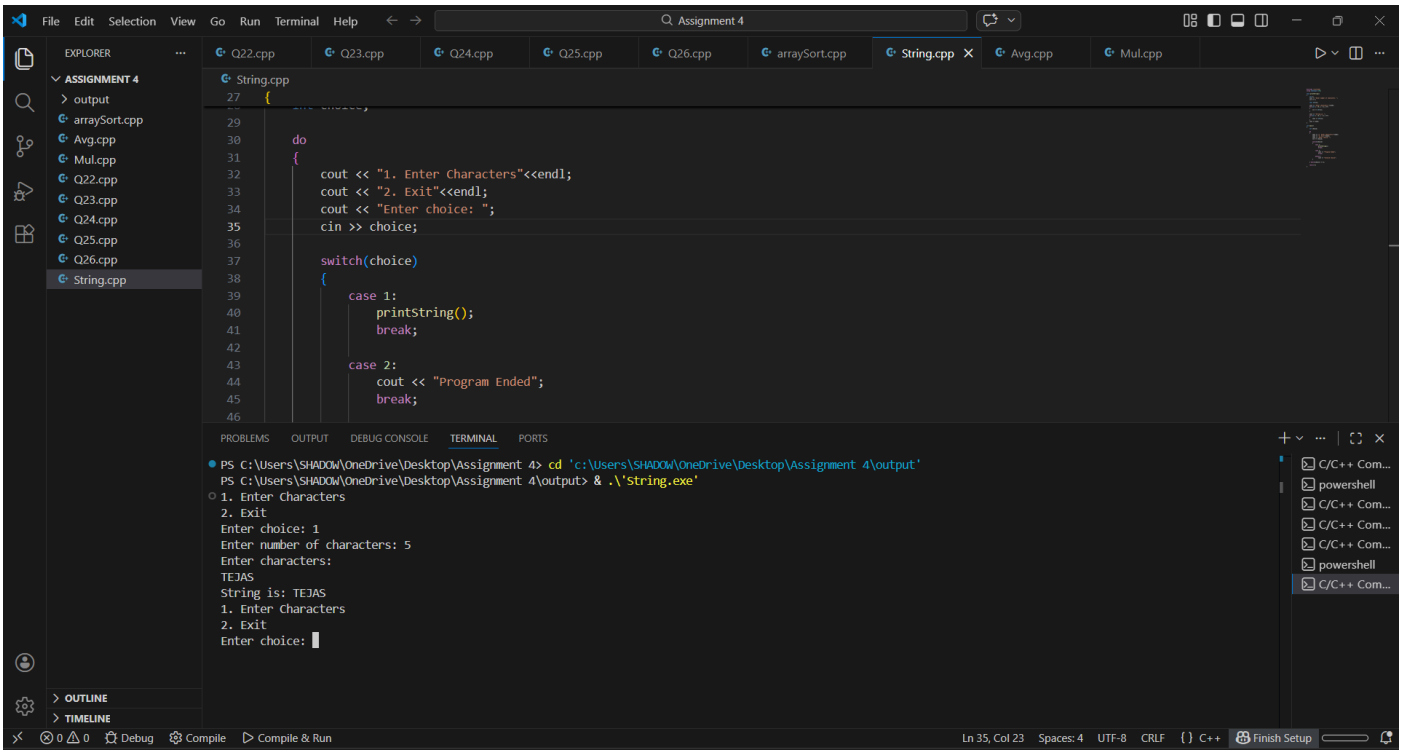
```
            break;

        case 3:
            delete[] arr;
            cout << "Program Ended"<<endl;
            break;

        default:
            cout << "Invalid choice"<<endl;
    }

  } while(choice != 3);

  return 0;
}
```



accept n characters from user and print is as string.

```
#include <iostream>
using namespace std;

void printString()
{
   int n;
   cout << "Enter number of characters: ";
   cin >> n;

   char str[n];

   cout << "Enter characters:"<<endl;
   for(int i = 0; i < n; i++)
   {
      cin >> str[i];
   }

   cout << "String is: ";
   for(int i = 0; i < n; i++)
   {
      cout << str[i];
   }
   cout << endl;
}

int main()
{
   int choice;

   do
   {
      cout << "1. Enter Characters"<<endl;
      cout << "2. Exit"<<endl;
```

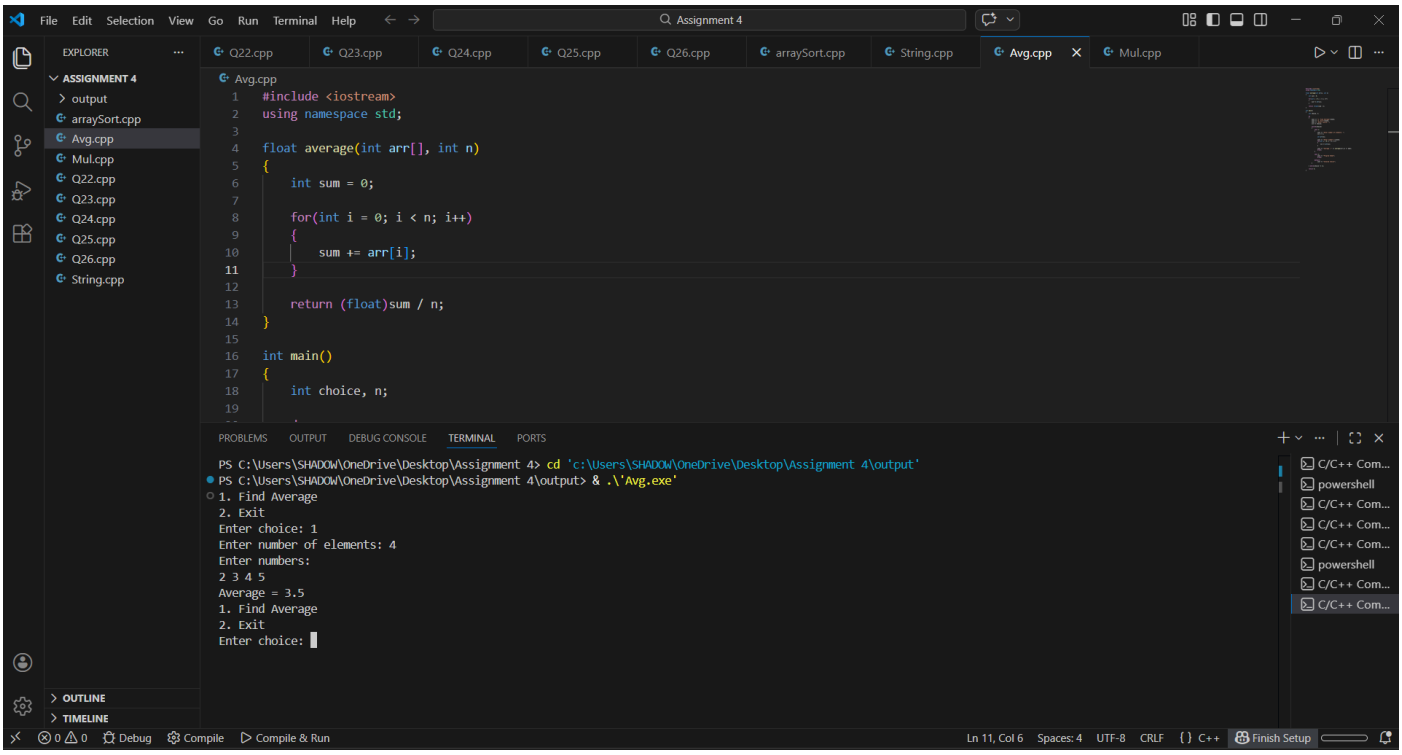
```
    cout << "Enter choice: ";
    cin >> choice;

    switch(choice)
    {
      case 1:
        printString();
        break;

      case 2:
        cout << "Program Ended";
        break;

      default:
        cout << "Invalid choice";
    }

  } while(choice != 2);
  return 0;
}
```



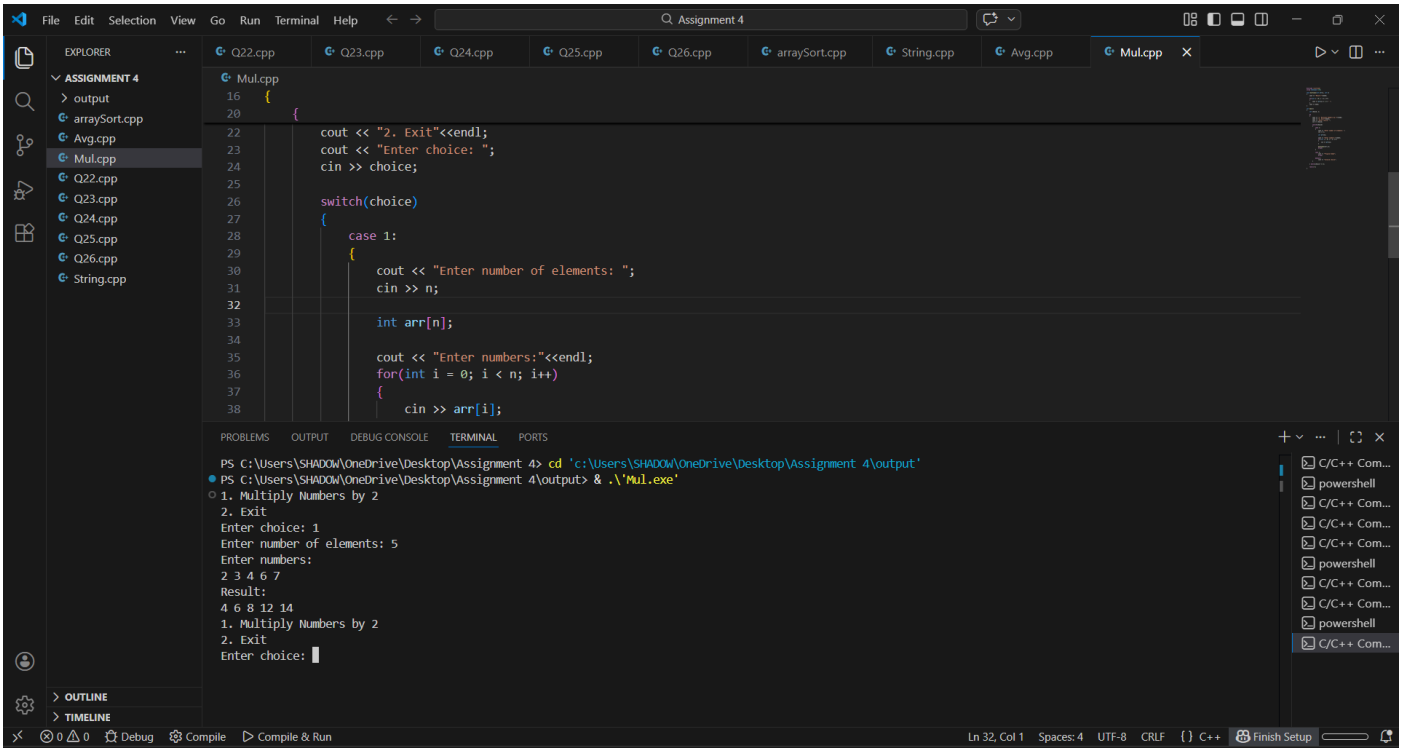
Accept n numbers from user and write a function to find out average and display average.

```
#include <iostream>
using namespace std;

float average(int arr[], int n)
{
   int sum = 0;

   for(int i = 0; i < n; i++)
   {
      sum += arr[i];
   }

   return (float)sum / n;
}

int main()
{
   int choice, n;

   do
   {
      cout << "1. Find Average"<<endl;
      cout << "2. Exit"<<endl;
      cout << "Enter choice: ";
      cin >> choice;

      switch(choice)
      {
         case 1:
         {
            cout << "Enter number of elements: ";
            cin >> n;
```

```
        int arr[n];

        cout << "Enter numbers:"<<endl;
        for(int i = 0; i < n; i++)
        {
            cin >> arr[i];
        }

        cout << "Average = " << average(arr,n) << endl;
        break;
    }
    case 2:
        cout << "Program Ended";
        break;
    default:
        cout << "Invalid choice";
    }
} while(choice != 2);
return 0;
}
```



Accept n numbers from user and write a function to multiply each number by 2 and display result.

```
#include <iostream>
using namespace std;

void multiply(int arr[], int n)
{
   cout << "Result:"<<endl;

   for(int i = 0; i < n; i++)
   {
      cout << arr[i] * 2 << " ";
   }
   cout << endl;
}

int main()
{
   int choice, n;

   do
   {
      cout << "1. Multiply Numbers by 2"<<endl;
      cout << "2. Exit"<<endl;
      cout << "Enter choice: ";
      cin >> choice;

      switch(choice)
      {
         case 1:
         {
            cout << "Enter number of elements: ";
            cin >> n;

            int arr[n];
```

```
        cout << "Enter numbers:"<<endl;
        for(int i = 0; i < n; i++)
        {
            cin >> arr[i];
        }

        multiply(arr,n);
        break;
    }
    case 2:
        cout << "Program Ended";
        break;
    default:
        cout << "Invalid choice";
    }
} while(choice != 2);
return 0;
}
```